

Pumping Lemma

The following conditions must be true:

- ◆ $|y| > 0$
- ◆ $|xy| \leq P$
- ◆ $x y^i z \in L$ for every $i \geq 0$

Question_1:

Using pumping lemma prove that $L = \{0^n 1^m 0^{nm} : n, m \geq 0\}$

$$S = 0^P 1^P 0^{P^2} \rightarrow P = 3 \rightarrow S = 0^3 1^3 0^9$$

$$S = 000111000000000 \quad (|x| = 1, |y| = 2, |z| = 12)$$

$$|y| > 0 \text{ and } |xy| \leq P \rightarrow |2| > 0 \text{ and } |1+2| \leq 3$$

$$\text{Follow rule } x y^i z \in L \text{ for } i = 2 \rightarrow S = x y^2 z$$

$$S = 00000111000000000 \quad (n = 5, m = 3, nm \neq 15) \rightarrow S \notin L$$

The string after pumping is not in the language $\rightarrow L$ is not regular

Question_2:

Using pumping lemma prove that $L = \{0^n 1^m : n < 3m\}$

$$S = 0^{3P-1} 1^P \rightarrow P = 3 \rightarrow S = 0^8 1^3$$

$$S = 0 \ 00 \ 00000 \ 111 \quad (|x| = 1, |y| = 2, |z| = 8)$$

$$|y| > 0 \text{ and } |xy| \leq P \rightarrow |2| > 0 \text{ and } |1+2| \leq 3$$

Follow rule $x y^i z \in L$ for $i = 2 \rightarrow S = x y^2 z$

$$S = 0 \ 0000 \ 00000 \ 111 \quad (n = 10, m = 3, n > 3m) \rightarrow S \notin L$$

The string after pumping is not in the language $\rightarrow L$ is not regular

Question_3:

Find the minimum pumping length for the language $1^*01^*01^*$

Result: $p = 3$

Shortest String : 00 (length 2)

Why $p \neq 2$:

For $s = 00$, any y we choose will be 0 and pumping it is not in the language (can't be pumped).

Why $p = 3$:

Any string $|w| \geq 3$ must contain an extra 1 that can be repeated without leaving the pattern.